Needle Detection in Interventional Pain Management with 3D Image Reconstruction

Essam A. Rashed, Mohammad al-Shatouri, Mona Selim and Hiroyuki Kudo

***Abstract*—Interventional pain management surgery is a clinical application performed though the guidance of x-ray fluoroscopy. Several protocols requires the injection of a needle close to the spinal cord to deliver a medication directly to the nerve system. The needle position information in the 3D space is important to avoid possible damage to the nerve system. It is common to perform the pain management surgery using C-arm scanner to follow up the treatment procedure. In many cases, it is difficult to observe the exact position of the injected needle through 2D images acquired using conventional C-arm scanner especially with complicated bone structures. It requires several attempts to image the patient from different positions and physician requires a mental process to imagine how the 3D structure looks like before starting the interventional procedure. This process may be repeated several times during a single interventional session, which cause a significant increase of radiation dose given to both patient and surgeon. In this paper, we introduce a method for needle detection in interventional pain management surgery using a clinical C-arm scanner. First, an in-house made gantry control unit (GCU) is mounted to the C-arm gantry to control the scanner orbital rotation. Second, the gantry rotation is traced using inertial measurement unit (IMU) sensor attached. A single cine loop is acquired by automatically rotate the C-arm gantry around the patient using GCU. Geometry information obtained from the IMU sensor is used to define the gantry position in the 3D space and synchronized with detector measurements in cine loop frames. The SCAN algorithm is then adopted for the 3D reconstruction of bone structures and injected needle.**

I. Introduction

Interventional pain management surgery is a clinical application performed though the guidance of x-ray fluoroscopy. Several protocols requires the injection of a needle close to the spinal cord to deliver a medication directly to the nerve system. The needle position information in the 3D space is important to avoid possible damage to the nerve system. It is common to perform the pain management surgery using C-arm scanner to follow up the treatment procedure. In many cases, it is difficult to observe the exact position of the injected needle through 2D images acquired using conventional C-arm scanner especially with complicated bone structures. It requires several attempts to image the patient from different positions and physician requires a mental process to imagine how the 3D structure looks like before starting the interventional procedure. This process may be repeated several times during a single interventional session, which cause a significant increase of radiation dose given to both patient and surgeon.

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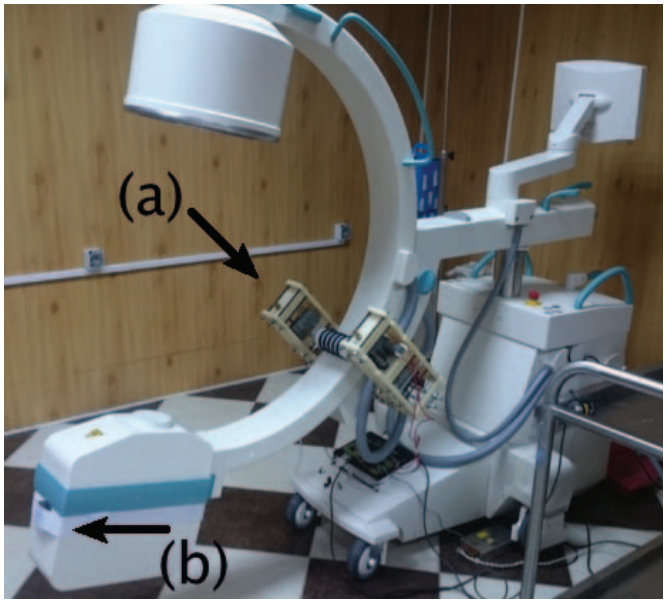

Fig. 1. C-arm scanner with (a) GCU and (b) IMU devices mounted.

In this paper, we introduce a method for needle detection in interventional pain management surgery using a clinical C-arm scanner. First, an in-house made gantry control unit (GCU) is mounted to the C-arm gantry to control the scanner orbital rotation [1]. Second, the gantry rotation is traced using inertial measurement unit (IMU) sensor attached as shown in Fig. 1. A single cine loop is acquired by automatically rotate the C-arm gantry around the patient using GCU. Geometry information obtained from the IMU sensor is used to define the gantry position in the 3D space and synchronized with detector measurements in cine loop frames. The SCAN algorithm [2] is then adopted for the 3D reconstruction of bone structures and injected needle.

II. Methods

A. SCAN algorithm

The developed system consists of three main components. The GCU, which controls the orbital movement of the C-arm scanner, the IMU sensor, which acquire the 3D position information, and the image reconstruction algorithm. We skip the details of the first two components and refer to reference [1] for more details. The image reconstruction problem can be presented as the solution of the set of linear equations:

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$$A\mathbf{x} = \mathbf{b}, \tag{1}$$

where $\mathbf{x} = (x_1, \ldots, x_n)$ is a vector representation of the 3D image, $\mathbf{b} = (b_1, \ldots, b_m)$ is vector of the detector measurements and A is the $m \times n$ matrix define the relation between x_j and b_i for all image voxels and detector bins. The imaging geometry of the application presented in this study is complicated as the data is highly limited due to the short gantry motion range. The SCAN algorithm [2] solves this problem by considering the following cost function:

$$\min_{(\mathbf{x},\mathbf{z})} f(\mathbf{x}) + g(\mathbf{z}) \quad s.t. \quad \mathbf{x} - \mathbf{z} = 0, \tag{2}$$

where $f(\mathbf{x})$ and $g(\mathbf{z})$ are defined by

$$f(\mathbf{x}) = \|\mathbf{x}\|_1^1, \tag{3}$$

$$g(\mathbf{z}) = \begin{cases} 0 & \text{if } A\mathbf{z} = \mathbf{b} \\ \infty & \text{(otherwise)} \end{cases}. \tag{4}$$

The alternating direction method of multipliers (ADMM) [3] is used to derive a solution of the cost function in Eq. (2) as follows:

$$\mathbf{x}^{(k+1)} = \arg\min_{\mathbf{x}} [f(\mathbf{x}) + \lambda^{(k)\top}(\mathbf{x} - \mathbf{z}^{(k)}) + \frac{\rho}{2}\|\mathbf{x} - \mathbf{z}^{(k)}\|_2^2],$$

$$\mathbf{z}^{(k+1)} = \arg\min_{\mathbf{z}} [g(\mathbf{z}) + \lambda^{(k)\top}(\mathbf{x}^{(k+1)} - \mathbf{z}) + \frac{\rho}{2}\|\mathbf{x}^{(k+1)} - \mathbf{z}\|_2^2],$$

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho(\mathbf{x}^{(k+1)} - \mathbf{z}^{(k+1)}),$$

where k is the iteration number and $\rho > 0$ is a weighting parameter of the second-order penalty term. The formulation of the cost function in Eq. (2) consider enforcing the sparsity of the scanned object through using the ℓ_1-norm. This approach is know to performs very well in sparse object (such as blood vessels) reconstruction from limited data. However, the problem investigated here is rather different as the object is non-sparse.

B. Sparcification through segmentation

The interesting observation of the application understudy is that the physician is interested only in high contrast objects such as bones and metal needle. These objects can be easily defined in the projection data domain through a segmentation process. Although, such process is known to reduce the image-data matching, the effect in the reconstructed image quality is rather small as the accurate intensity value of each voxel is not required for this application. It is enough to reconstruct the skeleton that represent the high contrast objects without intensity information that is required for CT diagnosis. The 2D projection data is segmented to remove low-contrast objects such as soft-tissue and preserve the high-contrast objects such as bone and metals. The following thresholding function is used for the segmentation process:

$$\tilde{b_i} = \begin{cases} b_i & (b_i \geq \epsilon) \\ 0 & (otherwise) \end{cases}, \quad i = 1, \ldots, m \tag{5}$$

where ϵ is a thresholding value define the structure of interest.

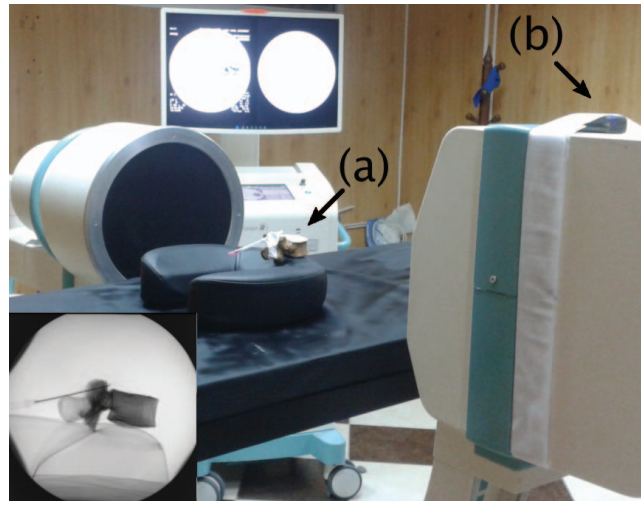

Fig. 2. Photograph of experiment setup using Ziehm Vision R C-arm scanner. (a) A cadaver vertebral bone with needle attached and (b) x-IMU sensor fixed with stripe tape to the x-ray tube. A 2D image obtained from the cine loop associated with the current position is shown in the bottom left corner.

III. Results and Discussion

To evaluate the performances of the proposed method, a lab experiment using real-data was performed. A cadaver vertebral bone is attached to a surgical needle and fixed with medical bandage tape as shown in Fig. 2(a). The cine loop is acquired using a single rotation orbit over 115° and 52 frames are extracted to be used for the reconstruction. A sample of the frames used in the reconstruction and the associated rotation angles are shown in Fig. 3. The C-arm scanner geometry is modeled for the 3D reconstruction similar to our previous work in [2]. The gantry position data extracted from the x-IMU sensor are synchronized with the cine loop frames. Reconstruction is obtained using 5 iterations and 3D volume rendering of the reconstructed vertebra is shown in Fig. 3. With the help of the 3D reconstructed volume, physician is being able to accurately determine the exact 3D position of the needle to avoid inaccurate positioning using mental processing of several 2D images. Although, some artifacts are observed in the 3D volume, the needle position can be verified accurately.

In this paper, we develop a new method for needle detection in interventional pain management surgery using a 3D reconstruction algorithm from a small number of views acquired over limited rotation orbit. The developed method requires a single cine loop data acquisition obtained using a C-arm scanner. The geometry data are synchronized with the cine loop frames and 3D image reconstruction is implemented. Results obtained from real data using cadaver vertebral bone with needle attached indicate that the proposed method is expected to have a practical usefulness in clinical surgery. In future, we would like to study the behavior of this method with real patient data.

References

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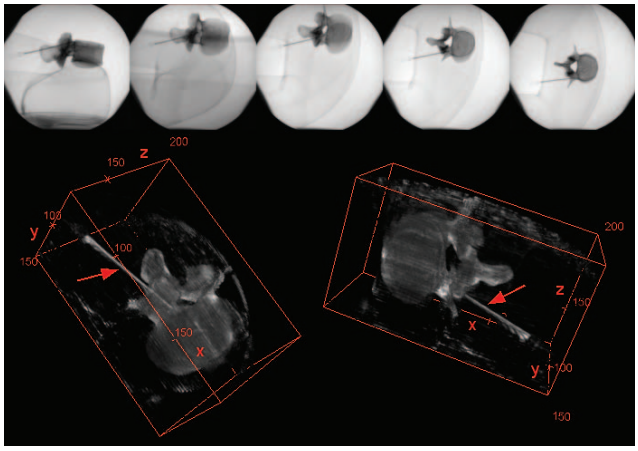

Fig. 3. Top is the cine loop frames corresponding to rotation angles of 9°, 31°, 53°, 75° and 97° (from left to right). Bottom are two positions of the 3D image and needle is marked with red arrow.

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